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Non-Parametrics

Contents

Parametrics Versus Non-Parametrics: Assumptions

Non-parametric statistics do not assume continuously scored, or normally distributed, dependent variable scores. By contrast, strictly speaking, parametric statistical analyses (e.g., *t*-test, ANOVA, regression, etc.) assume the dependent variable has been measured on an interval/ratio scale. Furthermore, strictly speaking, parametric statistics assume the data are distributed normally. However, as described many times across several chapters in this text, parametric statistics have been found to be remarkably robust to violations of both of these assumptions. For example, when dependent variables are measured on ordinal scales with at least 5-points or more, a parametric statistic can be expected to yield accurate *p*-value estimates, if the respondents used the entire scale. Additionally, when data are skewed less than |2.0| and kurtosis is less than |9.0|, parametric statistics can be expected to yield accurate *p*-value estimates, in many cases. Finally, when the data are severely non-normally distributed, bootstrapping has been found to be an attractive method to estimate standard errors, confidence intervals, and *p*-values. Given this knowledge, why should anyone ever conduct a non-parametric statistical analysis? Good question. As you'll see in this chapter, I have formed the opinion that there is probably always a better option than a non-parametric statistic, assuming the option is available to you (e.g., bootstrapping).

Many researchers believe that nonparametric statistics do not assume homogeneity of variance. Thus, non-parametrics should be viewed, advantageously, from this perspective. However, statistics such as the popular Mann-Whitney U statistic and the Kruskal-Wallis analysis are just as affected by violation of the assumption of homogeneity of variance (technically, assumption of homogeneity of distributional shapes) as the independent samples *t*-test and the between-subjects ANOVA (Zimmerman & Zumbo, 1993a). Consequently, one should not feel comforted when a non-parametric statistic result confirms a parametric statistic result. In all likelihood, the two statics are likely giving the same inaccurate result, when a shared assumption is violated. Furthermore, an abundant amount of simulation research has found the *t*-test and the ANOVA to be robust to violations of the homogeneity of variance assumption, when the sample sizes are equal. When the sample sizes are unequal, there are very attractive parametric statistics such as Welch's *t*-test and the Brown-Forsythe adjusted *F*-test, the Games-Howell multiple comparison procedure, and the Huyhn-Feldt adjusted repeated measures ANOVA. In light of the above, researchers should find themselves in a position to use parametric statistics justifiably to test their hypotheses, in the vast majority of cases.

As a general statement, keeping within a parametric statistical framework is a good thing for two reasons. First, it allows the researcher to make inferences about the data they collected, rather than data that have been ranked (most non-parametric statistical techniques transform the raw data into ranks). Secondly, I believe most researchers have in mind hypotheses that are more consistent with parametric statistical analyses, rather than the hypotheses that underlie non-parametric statistical analyses.

Despite the above, there are some good reasons to have knowledge of commonly applied parametric statistics. First, when the dependent variable has been measured on a 4-point scale or less, or a ranking measurement method has been applied, and bootstrapping is not available, a non-parametric statistic may be your best option. Relatedly, when the data are severely non-normally distributed (skew > 2.0), and access to a bootstrapping program is not available, a non-parametric statistic may be the best option. Thirdly, there is evidence to support the notion that nonparametric statistics can be more powerful than parametric statistics, in some cases (Zimmerman & Zumbo, 1993a). Fourthly, non-parametric statistics are commonly reported in scientific reports. Consequently, knowledge of nonparametric statistical analyses is useful, although perhaps not as useful as many may believe. Finally, your colleague or supervisor might not believe you, when you try to tell them about the information you learned in this chapter, so, you might get stuck conducting a non-parametric statistic, even though you know there are better options. (**Watch Video 16.1:** Reasons for Non-Parametric Statistics).

Ordinal Dependent Variables

Ordinal dependent variables are fairly commonly employed by researchers, especially those who use survey/questionnaire type instruments. The Likert scale is an ordinal scale. For example, a personality questionnaire may include the following item: “I like to have routine in my life” and the participants respond to the item on the following scale: 1 = strongly disagree, 2 = disagree, 3 = neither agree nor disagree, 4 = agree, and 5 = strongly agree. Typically, researchers sum the responses across several items measured on a Likert scale into a total score. Such a sum score would be associated with an interval scale. Consequently, in this case, parametric statistics should be applied to analyses that include the total score as a dependent variable. However, if a researcher wished to test hypotheses at the item level, or perhaps the researcher administered a single item as a dependent variable, then non-parametric analyses may be required.

Simulation research has shown that parametric tests (Pearson correlation; *t*-test, etc.) will work in cases with an ordinal dependent variable, so long as the scale has at least 5-points

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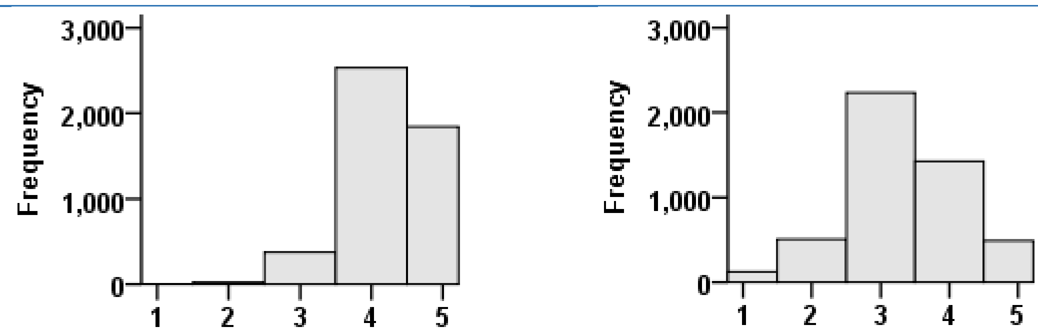
(de Winter & Dodou, 2010; Rasmussen, 1989). However, a major qualifier over such a recommendation needs to be considered: the participant responses associated with the 5-point scale need to cover the whole spectrum of the scale. That is, all too often, a theoretical 5-point scale actually turns out to be a 3-point scale. For example, I am familiar with a questionnaire that has the following item: 'I create a positive work environment for others'. Participants respond to the item based on the following 5-point scale: 1 = Almost Never; 2 = Seldom; 3 = Sometimes; 4 = Usually; 5 = Almost Always. Based on a sample of 4,775 participants, the histogram depicted in Figure C16.1 (Panel A) was obtained (**Watch Video 16.2:** 5-Point Histogram & Non-Parametrics).

Clearly, this theoretical 5-point scale was really only a 3-point scale in "reality". By comparison, another item from the same questionnaire ('I take criticism from colleagues personally.') yielded data that may be regarded as genuinely consistent with a 5-point scale. Ultimately, the onus should be on the researcher to demonstrate that the full 5-point scale has been used by the participants, prior to applying parametric statistics. If the item depicted in Panel A were used as a dependent variable, it would not be appropriate to perform a parametric statistical analysis, as the analysis would not yield a trustworthy standard error or *p*-value. Instead, it would be necessary to perform a nonparametric analysis (or perhaps bootstrapping).[1]

Figure C16.1. Histogram of 5-Point Scale

Panel A | Panel B



Mann-Whitney U: Difference Between Two Mean Ranks

Adan and Natale (2002) were interested in examining the difference between males and females in their morningness and eveningness preferences. Consequently, they administered the Morningness-Eveningness Questionnaire to 2135 adults (51.2% female). The

[1] Of course, some researchers may be interested in applying polychoric correlations with the 5-point Likert data discussed, here, which may be the best approach, depending on the research question (see Chapter 5).

MEQ consists of 19 items, most of which are scored on 4-point scale. Amongst other items, Adan and Natale (2002) tested the difference between males and females on item 16: Desire for physical exercise between 10pm to 11pm. The 4-point response scale consisted of: 1 = Would be on good form, 2 = Would be on reasonable form, 3 = Would find it difficult, 4 = Would find it very difficult. As the dependent variable was an ordinal scale with less than 5-points, it was necessary to perform a nonparametric statistic on the data to test differences between males and females.

The Mann-Whitney test is similar to the independent samples *t*-test in that it can test the difference between two groups with respect to central tendency. However, it should be made clear that the Mann-Whitney test is a test of the difference between mean ranks. The Mann-Whitney test is not a test of the difference between means. It also not a test of the difference between medians (as many would lead you to believe). Technically, the Mann-Whitney test follows the *z*-distribution (not the *t*-distribution like the independent samples *t*-test). In this example, a Mann-Whitney test would be appropriate, as there were only two groups (males and females). I have simulated some data to correspond very closely to the results reported by Adan and Natale (2002) (**Data File: MEQ_data**).

Mann-Whitney U: SPSS

Operationally, the Mann-Whitney test converts the ordinal data into ranked data, and then the statistical analysis is performed on the ranked data. To conduct a Mann-Whitney analysis in SPSS, one can use the 'Non Parametric Tests' menu option in SPSS (**Watch Video 16.3:** Mann Whitney U Test in SPSS).

As can be seen in the SPSS table entitled 'Ranks', the males were associated with a mean rank of 954.46 and the females were associated with a mean rank of 1176.04. Thus, at least numerically, the females, on average, scored higher than the males. A higher scores suggests that the females are more tired in the evening.

Ranks

	gender	N	Mean Rank	Sum of Ranks
Desire for physical exercise between 10.00pm and 11.00pm	Male	1041	954.46	993594.50
	Female	1094	1176.04	1286585.50
	Total	2135		

Next, SPSS produced a table entitled 'Test Statistics' which includes the key results associated with the Mann-Whitney test. Ignore the Mann-Whitney U value (451233.500), as well as the Wilcoxon *W* value (993594.50). The focus should be on the *Z*-value and the *p*-value (Aysmp. Sig.(2-tailed). In this case, the null hypothesis of equal mean ranks has been rejected, $Z = -8.78$, $p < .001$. Thus, females, on average, are more tired than males for the purposes of engaging in physical activity in the late evening.

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Test Statistics[a]

	Desire for physical exercise between 10.00pm and 11.00pm
Mann-Whitney U	451233.500
Wilcoxon W	993594.500
Z	-8.776
Asymp. Sig. (2-tailed)	.000

a. Grouping Variable: gender

Effect Size

Although not well known, a measure of effect size can be calculated for the Mann-Whitney statistic. The effect size estimate is known as eta-squared (η^2), because it represents the percentage of variance in the ranks that was accounted for by the grouping (independent) variable. The formula corresponds to:

$$\eta^2 = \frac{Z^2}{N-1} \tag{1}$$

$$\eta^2 = \frac{-8.78^2}{2135} = \frac{77.09}{2135} = .0361$$

Thus, 3.61% of the variance in the ranks was accounted for by sex, which would be close to a medium effect, based on Cohen's (1988; 1992) guidelines. Typically, researchers prefer to report effect sizes as Cohen's *d* in the context of the difference between two groups. Cohen's *d* can be estimated from a Mann-Whitney analysis with following formula:

$$d = Z * \sqrt{\frac{1}{N_1} + \frac{1}{N_2}} \tag{2}$$

$$d = -8.776 * \sqrt{\frac{1}{1041} + \frac{1}{1094}}$$

$$d = -8.776 * \sqrt{.00096 + .00091}$$

$$d = -8.776 * .0432$$

$$d = -.379$$

Thus, the male mean ranking (954.46) on the item was approximately 38% of a standard deviation lower than the female mean ranking (1176.04). Such an effect would be considered close to a medium effect, based on Cohen's (1988; 1992) guidelines.

Mann-Whitney: Assumptions

I often hear researchers comfort themselves by saying that they were a bit worried that their data may have violated a particular assumption, but because the corresponding non-parametric statistic corroborated the *p*-value result obtained from a parametric statistic, everything should be fine. For example, a researcher may conduct a Mann-Whitney test to corroborate a statistically significant *p*-value obtained from an independent samples *t*-test. Unfortunately, it is a common misconception that non-parametric statistics such as the Mann-Whitney do not have assumptions. In actual fact, non-parametric statistics have essentially the same assumptions as parametric statistics, with the exception of normally distributed dependent variables (Maxwell & Delaney, 1990). For example, the Mann-Whitney test certainly does assumes homogeneity of variance. More specifically, the Mann-Whitney test assumes that the dependent variable scores are distributed in the same way across groups (i.e., non-normal in the same way; Wilcox, 2003). One way to test whether the distributions are similar across groups is with a homogeneity of variance test, because the observation of statistically significant different variances necessarily implies different distributional shapes (Maxwell & Delaney, 1990). Thus, in a large number of cases, researchers are likely obtaining similar results between parametric and non-parametric results, because both the parametric and nonparametric statistics are inaccurate in the same way.

Unfortunately, most programs do not test homogeneity of variance when a Mann-Whitney statistics is performed. However, the assumption can be tested using different utilities in SPSS.

Homogeneity of Variance for Mann-Whitney: SPSS

One option to test the assumption of homogeneity of variance in SPSS in the Mann-Whitney case is to use the ‘Explore’ menu option in SPSS (**Watch Video 16.4:** Robust (Median) Levene's Test in SPSS). Amongst other output, SPSS will produce the following table entitled ‘Test of Homogeneity of Variance’. In this case, focus should be placed on the ‘Based on Median and with adjusted df’ row of results, which correspond to the non-parametric Levene’s test of homogeneity of variance. In this example, the assumption of equal variances was not violated, based on the Levene’s homogeneity of variance test, $F(1, 2132.67) = .57$, $p = .449$. Because the assumption was not violated, the Mann-Whitney test results can be interpreted. Had the assumption been violated, the *p*-value would have been inaccurate.

Test of Homogeneity of Variance

		Levene Statistic	df1	df2	Sig.
Desire for physical exercise between 10.00pm and 11.00pm	Based on Mean	.362	1	2133	.547
	Based on Median	.574	1	2133	.449
	Based on Median and with adjusted df	.574	1	2132.667	.449
	Based on trimmed mean	.471	1	2133	.493

Mann-Whitney on Original Ranks: Example 2

In some cases, as above, a researcher will have a dependent variable measured on a limited information ordinal scale (less than 5-points), which may require use of a nonparametric statistic such as the Mann-Whitney to test a hypothesis relevant to the difference between two groups. In such cases, the Mann-Whitney procedure will transform the ordinal data into ranks, prior to conducting the statistical analysis. In other cases, however, the dependent variable may be measured with ranks to start with. For example, the participants may have been asked to rank particular items in order of preference. In such cases, the Mann-Whitney procedure would still be appropriate, if the hypothesis relates to the difference between two independent groups, as the Mann-Whitney will convert data into ranks, whether they were ranked to begin with or not.

One such case is a study by Perilloux, Fleischman, and Buss (2011). Amongst other things, they were interested in testing the difference in potential mate characteristic preferences between men and women. Consequently, they had a group of men (N = 96) and women (N = 213) rank their preference for 13 mate characteristics, including such things as intelligent, attractive, creative, earning capacity, and personality, for example. I have simulated some data to correspond very closely to the results reported by Perilloux, Fleischman, and Buss (2011) with respect to the earning capacity (**Data File: mating_earning_capacity**). In this case, the null hypothesis corresponded to: Males and females rank earning capacity as equally attractive in a potential mate. The Mann-Whitney test can test such a null hypothesis (**Watch Video 16.5:** Mann-Whitney U on Ranks in SPSS).

In SPSS, the analysis can be performed in the exactly the same manner as the preceding example. As can be seen in the SPSS table entitled ‘Ranks’, the males were associated with a mean rank of 87.65 and the females were associated with a mean rank of 185.35. Given that the participants provided only ranks between 1 and 13, such mean ranks are not readily interpretable. What has happened is that SPSS has re-ranked the data across both the males and females, consequently, the rank values have taken on values much greater than 13, as described in the video.

Ranks

	sex	N	Mean Rank	Sum of Ranks
earning_capacity	Males	96	87.65	8414.50
	Females	213	185.35	39480.50
	Total	309		

Next, SPSS reported the results associated with the Mann-Whitney test in the 'Test Statistics' table. The null hypothesis has been rejected, Z = -9.00, p < .001. Thus, on average, females ranked earning capacity in a mate more highly than males. In the Advanced Topics section of this chapter, I deal with the fact that these data violated the assumption of homogeneity of variance (and, therefore, homogeneity of distributional shapes).

Test Statistics[a]

	earning_capacity
Mann-Whitney U	3758.500
Wilcoxon W	8414.500
Z	-9.002
Asymp. Sig. (2-tailed)	.000

a. Grouping Variable: sex

Again, I calculated an estimate of effect size based on the η^2 formula presented in the first example. In this example, η^2 was estimated at .263. Thus, 26.3% of the variability in the ranks was accounted for by sex.

$$\eta^2 = \frac{-9.00^2}{309-1} = \frac{81}{308} = .263$$

Wilcoxon Signed-Rank Test versus Mann-Whitney

It is important to note that Wilcoxon also developed a Sign-Ranked Test for independent groups (Wilcoxon, 1945), which may be considered a competitor to the Mann-Whitney test. However, it never really became popular, unlike Wilcoxon's within-subjects test of the difference between two related conditions (described later in this chapter). It turned out that Mann-Whitney's test (Mann & Whitney, 1947) gives the same result as Wilcoxon's test of the difference between two independent groups. For this reason, the two tests are sometimes referred to as the Wilcoxon-Mann-Whitney test.

Kruskal-Wallis: Difference between Two or More Mean Ranks

If you haven't already, you will likely find yourself one day at the dining table with a child refusing to eat his/her vegetables. What should you do? Pay your child to eat them? Cooke et al. (2011) addressed this question. To do so, Cooke et al. (2011) assigned children (4 to 6 years old) randomly to four groups: exposure to vegetables with tangible rewards

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(stickers), exposure to vegetables with social rewards (praise), exposure to vegetables with no rewards, and a control group. Then, all of the children tasted vegetables 12 days in a row (except the control group). Finally, a day after the last day of tasting, Cooke et al. (2011) had the children rate the degree to which they liked vegetables on a 3-point scale: 1 = 'yucky'; 2 = 'just OK'; 3 = 'yummy'. Cooke et al. (2001) hypothesized that the level with which children liked vegetables after the 12 tasting sessions would not be equal across the groups. Correspondingly, the null hypothesis was that the level with which children liked vegetables after the 12 tasting sessions would be equal across the groups.

Because the dependent variable in the Cooke et al. (2011) investigation was clearly ordinal in nature, parametric statistics are not an acceptable option. Instead, a non-parametric statistic needs to be applied. As the design was a between-subjects design and there were more than two groups, the Kruskal-Wallis statistic is an acceptable option to test the null hypothesis that groups were associated with equal mean ranks.

In a manner similar to the Mann-Whitney test statistic, the results associated with a Kruskal-Wallis statistic are relevant to mean ranks, because the dependent variable data are converted to ranks in a Kruskal-Wallis. Thus, the Kruskal-Wallis statistic does not test the difference between means. It tests the difference between mean ranks.

I have simulated some data to correspond to the results reported by Cooke et al. (2011). Below are the bar charts associated with the three types of measures of central tendency: mean ranks, medians, and means (**Data File: veggies_oneway_between**)

As can be seen in panel A (Figure C16.2), the mean ranks were such that the control group had the lowest numerical mean rank (155.20). Additionally, the tangible reward and social praise conditions were associated with very similar mean ranks (243.26 and 241.97, respectively). Finally, the exposure only group was associated with a mean rank of 210.85. As I demonstrate below, the fact that the three exposure groups were all associated with the same median will help prove that tests such as the Kruskal-Wallis (and the Mann-Whitney) do not have anything to do with medians. Finally, in panel C, it can be seen that the means exhibited a pattern very similar to the mean ranks. In order to test the hypothesis that all four groups were associated with equal mean ranks, a Kruskal-Wallis analysis was performed.

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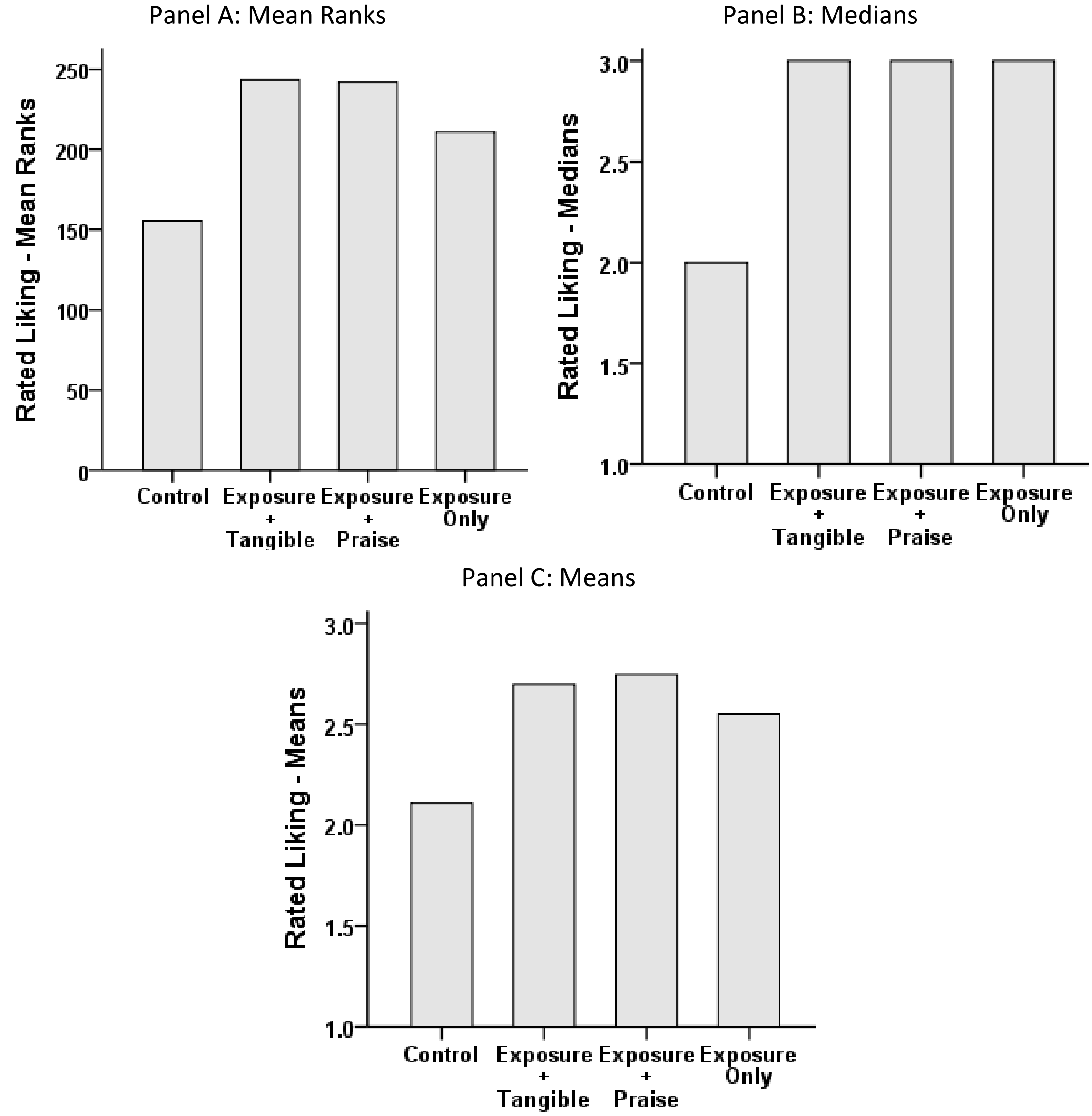
Figure C16.2. Bar Charts of Measures of Central Tendency



Kruskal-Wallis: SPSS

There are two options within the SPSS menus to perform a Kruskal-Wallis. I prefer the option through the legacy dialogs (**Watch Video 16.6:** Kruskal-Wallis in SPSS). As can be seen in the SPSS table entitled 'Ranks', the control group was associated with the smallest numerical mean rank (*M* = 155.20). By contrast, the Exposure + Tangible group was associated with the largest numerical mean rank. Of course, it is necessary to determine whether the numerical difference in the mean ranks reported in the table entitled 'Ranks' were statistically significantly different from each other.

Ranks

	group	N	Mean Rank
rated_liking	Control	112	155.20
	Exposure + Tangible	99	243.26
	Exposure + Praise	106	241.97
	Exposure Only	105	210.85
	Total	422	

As can be seen in the SPSS table entitled 'Test Statistics', the null hypothesis of equal mean ranks was rejected, $\chi^2(3) = 52.13$, $p < .001$.

Test Statistics[a,b]

	rated_liking
Chi-Square	52.131
df	3
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable: group

Effect Size

Although not commonly observed in the literature, η^2 can be calculated in the Kruskal-Wallis case based on the following formula:

$$\eta^2 = \frac{\chi^2}{N-1} \qquad (3)$$

$$\eta^2 = \frac{52.131}{422-1} = .124$$

Thus, 12.4% of the variance in vegetable rating ranks was accounted for by group membership. Such an effect would be considered large, based on Cohen's (1988; 1992) guidelines. Because the Kruskal-Wallis is an omnibus statistic, just like the one-way between-subejcts ANOVA, it is necessary to follow-up the statistically significant Kruskal-Wallis with follow-up multiple comparisons, as it is not known which mean ranks were statistically significantly different from each other.

Kruskal-Wallis: Follow-Up Multiple Comparisons

A statistically significant Kruskal-Wallis simply indicates that there is at least one group comparison that is statistically significant. However, in this case, the Kruskal-Wallis may be considered an omnibus statistic, just like the one-way between-subjects ANOVA. There are no well-established non-parametric multiple-comparison procedures for analyses based on ranks. Consequently, in order to protect oneself against familywise error inflation, the only viable option is to perform a series of Mann-Whitney analyses with the implementation of a Bonferroni correction. Typically, researchers divide the per contrast alpha (i.e., .05) by the

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number of comparisons performed, in order to keep the familywise error rate at .05. In this example, in order to keep the familywise error rate at .05, each contrast would have to be tested against alpha = .008. However, the logic of the Bonferroni adjustment extends to the multiplication of each estimated *p*-value by the number of comparisons performed. The *p*-values that remain less than .05, after the adjustment, would be declared statistically significant (**Watch Video 16.7:** Kruskal-Wallis Multiple Comparisons in SPSS).

The results associated with the six pairwise Mann-Whitney multiple comparisons are reported in Table C16.1. I have reported the unadjusted *p*-values (*p*) and the Bonferroni adjusted *p*-values (*p*'). It can be observed that all of the comparisons with the control group were significant statistically ($p < .05$). However, there were no significant differences between the treatment groups. Keeping mind that it is difficult to support the null hypothesis, it may be suggested that simply exposing children to the vegetables 12 times (Exposure Alone) helped increase their liking of them in a manner very similar to the conditions which included rewards (tangible and social). The mere-exposure effect strikes again!

Table C16.1. Summary of Multiple-Comparisons

			Z	p	p'	η^2
Control	vs	ETR	-5.60	< .001	< .001	.149
Control	vs	EP	-6.00	< .001	< .001	.165
Control	vs	EA	-3.97	< .001	< .001	.073
ETR	vs	EP	-.39	.695	.999	.001
ETR	vs	EA	2.57	.010	.060	.032
EP	vs	EA	-2.48	.013	.078	.029

Note. ETR = exposure plus tangible rewards; EP = exposure plus praise; EA = exposure alone; *p*' = Bonferroni adjusted *p*-value.

Kruskal-Wallis: Homogeneity of Variance

As I demonstrated above, Kruskal-Wallis and ANOVA on the ranks tend to yield identical results, because they are essentially identical tests. Given the close similarity between Kruskal-Wallis and a parametric statistic such as ANOVA, it may not come as a surprise to you that Kruskal-Wallis essentially assumes homogeneity of variance. Specifically, although Kruskal-Wallis does not assume normality, it does assume that the dependent variable distributions associated with the groups are the same. That is, Kruskal-Wallis assumes homogeneity of distributional shapes. So, the distributions can be non-normal, but they should be so in the same way (Wilcox, 2003). One way to test whether distributions are similar is with a homogeneity of variance test, because the observation of statistically significant different variances necessarily implies different distributional shapes (Maxwell & Delaney, 1990).

In this example, the assumption of equal variances was violated, based on Levene's homogeneity of variance test on ranks, $F(3, 48) = 21.04$, $p < .001$. However, it should be acknowledged that the sample sizes were large across the groups. Consequently, even small discrepancies in variances would be expected to be detected as statistically significant. Nonetheless, in the Advanced Topics section of this chapter, I describe how the Brown-Forsythe F-test can be applied in this case to test the null hypothesis of equal mean ranks, when the homogeneity of variance assumption has been violated.

Test of Homogeneity of Variances

rated_liking_ranks

Levene Statistic	df1	df2	Sig.
21.044	3	418	.000

Within-Subjects: Two-Levels – Wilcoxon Signed-Rank Test

Most people tend to overestimate their abilities, including medical doctors. Barnsley et al (2004) had 30 junior medical doctors rate their competence across seven medical procedures, one of which was cardiopulmonary resuscitation (CPR). Barnsley et al (2004) also had several senior medical doctors assess the competence of the same 30 medical doctors across the seven medical procedures, again, including CPR. The level of measurement associated with the collected data was limited ordinal, as the scale was associated with only four points, as can be seen in Table 16.2 (**Data File: doctors_confidence**).

With respect to CPR, the junior doctors rated their confidence at $M = 2.53$ ($SD = 0.90$). By contrast, on the comparable 4-point competence scale, the senior doctors rated the junior doctors' competence at $M = 1.27$ ($SD = 0.52$). Clearly, there was a fairly substantial numerical difference. Was the difference statistically significant? Because the dependent variable was measured on a limited information ordinal scale (i.e., less than 5-points), a non-parametric statistic should be considered in this case. As there were only two levels in the independent variable (junior confidence vs. senior rated competence), the Wilcoxon Signed-Rank Test was a good option to test the hypothesis.

The Wilcoxon Signed-Rank Test is similar to the paired samples t-test in the sense that it tests the difference between two within-subjects conditions. The main difference is that, in the Wilcoxon Signed-Rank Test case, the data are ranked, if they are not already in rank form. Additionally, the Wilcoxon Signed-Rank Test does not calculate a standard error of the difference between means, or mean ranks, or medians for that matter. Many researchers appear to be under the mistaken impression that the Wilcoxon Signed-Rank Test is a test of the difference between two medians. It is not. Statistical tests of the difference between medians have proven relatively difficult to generate and establish (Bonnett & Price, 2002; Wilcox, 1991; Wilcox, 2006). Very specifically, the Wilcoxon Signed-Rank Test is a test of the difference between the summed deviations ranked (signed negative or positive) versus the null hypothesis of zero, based on paired data.

Table C16.2. Scales Used by Doctors to Rate Confidence and Competence

	Confidence		Competence
1	Not yet confident to do unsupervised.	1	Not yet competent to do unsupervised.
2	Fairly confident to perform the procedure without supervision.	2	Fairly competent to perform the procedure without supervision.
3	Confident to perform the procedure without supervision.	3	Competent to perform the procedure without supervision.
4	Confident to teach the procedure.	4	Competent to teach the procedure.

Note. Junior doctors rated their confidence; senior doctors rated the junior doctors' competence.

It is important to keep in mind that, in contrast to between-subjects designs, the data are ranked within cases in the Wilcoxon Signed-Rank Test, not across cases. Thus, as the Wilcoxon Signed-Rank Test can accommodate only two levels (e.g., time 1 and time 2), each cases data are ranked 1 and 2.

It should also be noted that, in most applications, the paired data in a within-subjects design will be obtained from the same participants in a typical time 1/time 2 scenario. However, there are within-subjects designs in which participants supply information on only one occasion. For example, this confidence versus competence investigation. The data are paired, which makes it a within-subjects design, irrespective of the fact that the participants, in this case, supplied only one column of data (**Watch Video 16.8:** Two Types of Within Subjects Designs).

Wilcoxon Signed-Rank Test - SPSS

In order to conduct a Wilcoxon Signed-Rank Test in SPSS, one can use the 'Nonparametric Tests' menu option (**Watch Video 16.9:** Wilcoxon Signed-Rank Test in SPSS). As can be seen in the SPSS table entitled 'Ranks', there were five negative ranks, 0 positive ranks, and 5 ties. Thus, a total of 25 junior medical doctors rated their confidence levels higher than the senior doctors rated their competence, and five junior doctors rated their confidence levels equal to their corresponding competence levels. Why SPSS reports this information for this analysis is not entirely clear to me, as it is more directly relevant to another statistical analysis known as the 'sign test'.

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Ranks

		N	Mean Rank	Sum of Ranks
competence - confidence	Negative Ranks	25[a]	13.00	325.00
	Positive Ranks	0[b]	.00	.00
	Ties	5[c]		
	Total	30		

a. competence < confidence

b. competence > confidence

c. competence = confidence

Next, the key result associated with the analysis is reported in a SPSS table entitled 'Test Statistics'. It can be seen that the null hypothesis of equal mean ranks was rejected, Z = -4.48, p < .001.

Test Statistics[a]

	competence - confidence
Z	-4.481[b]
Asymp. Sig. (2-tailed)	.000

a. Wilcoxon Signed Ranks Test

b. Based on positive ranks.

Effect Size

Unfortunately, SPSS does not calculate an effect size estimate in the Wilcoxon-Signed-Rank Test case. However, an approximate effect size estimate can be calculated with the following formula:

$$\text{partial}\,\eta^2 = \frac{Z^2}{(N(k-1)} \tag{4}$$

Where Z^2 = the Z-value test statistic (squared); N = total number of pairings; and k = number of levels (k is always equal to 2 in the Wilcoxon Sign-Rank Test)

$$\text{partial}\,\eta^2 = \frac{-4.481}{(30(2-1)} = \frac{20.07}{30} = .669$$

Thus, 66.9% of the variance in the ranks was accounted for by the independent variable, controlling for the shared variance between the confidence and competence ratings. I used the word "approximate" above, because I do not believe the results associated with a Wilcoxon Sign-Rank Test can be described in such a way, exactly. For this reason, I prefer not to use it. Instead, I prefer to use the Friedman's ANOVA, as described next.

Friedman's ANOVA: Within-Subjects Mean Ranks – Ranked Data

I once came across a blog post entitled, "Is wine bullshit?" I was intrigued, because I have often wondered myself, although not in those precise words. Can people really tell the difference between a cheap wine and an expensive wine? Some studies say 'Yes", and others say, 'No', although none of the studies are especially compelling. One study demonstrated that wine drinkers can be easily duped into thinking they are drinking an expensive wine, when they are not (Morrot, Brochet, & Dubourdieux, 2001). In my mind, the observation that wine drinkers can be so easily manipulated, when it comes to their evaluations of the taste of wines, suggests that if there is a "true" difference in the perceived quality of the taste of wines, it is likely a weak effect. But who knows with a good level of confidence?

One way this "bullshit" question could be answered would be to purchase a variety of 10 variously priced wines. Then, have 20 wine connoisseurs rank individually each of the 10 wines, based on their individual tastings of the wines. A rank of 1 would imply the best tasting wine and a rank of 10 would imply the worst tasting wine. Because each of the 10 wines would be tasted by the same 20 connoisseurs, it would be a within-subjects design. Because the dependent variable would be ordinal in nature (ranks), a non-parametric statistic should be applied. In this case, Friedman's ANOVA would be a good candidate to test the null hypothesis of equal mean ranks.

Many researchers (even statisticians) are under the impression that Friedman's ANOVA tests a hypotheses relevant to medians. I have never seen this demonstrated, and I'm pretty sure it does not. Instead, like the Kurskal-Wallis, Friedman's ANOVA tests a hypotheses relevant to mean ranks. Friedman's ANOVA is a within-subjects design analysis, and it can handle more than two levels. Consequently, it may be considered the non-parametric analogue of the oneway within-subjects ANOVA. In contrast to the within-subjects ANOVA, Friedman's ANOVA is commonly claimed not to assume sphericity. However, my impression is that it would be more accurate to suggest that the ranking process employed at step 1 of the Friedman's ANOVA procedure eliminates the possibility of violating the assumption of sphericity.

Hypotheses

Null Hypothesis (H_0): The 10 wines will be associated with equal taste preference mean ranks.

Alternative Hypothesis (H_1): The 10 wines will be associated with unequal taste preference mean ranks.

Thus, if the 20 wine connoisseurs exhibit at least some of their supposed skill at identifying good wines from bad, at least one bottle of wine will be associated with a disproportionate number of low or high ranks. In this example, the data were ranked by the participants. Consequently, the Friedman's ANOVA procedure may be considered appropriate to test the null hypothesis.

Friedman's ANOVA: SPSS

Unfortunately, I could not find a real study to base the results upon. Perhaps someone reading this will be inspired to carry out the investigation, one day. Nonetheless, I have simulated some data to help demonstrate the utility of Friedman's ANOVA (**Data File: wine_testing**). In order to conduct a Friedman's ANOVA, you can use the 'Nonparametric Tests' menu option in SPSS (**Watch Video 16.11:** Friedman's ANOVA on Ranked Data in SPSS).

The first table outputted by SPSS is the entitled 'Descriptive Statistics'. It can be observed that the means were not numerically equal. In this case, the means actually correspond to the mean ranks, but this will not always be the case (i.e., if the raw data were ordinal scaled data rather than ranks). The medians were also not equal numerically. However, as mentioned above, and demonstrated in the second example below, Friedman's ANOVA is not relevant to medians.

Descriptive Statistics

						Percentiles		
	N	Mean	Std. Deviation	Minimum	Maximum	25th	50th (Median)	75th
wine1	20	5.00	2.884	1	10	2.25	5.00	7.75
wine2	20	5.90	3.093	1	10	3.25	7.00	9.00
wine3	20	5.05	2.762	1	10	2.25	5.50	7.00
wine4	20	7.25	2.314	3	10	5.25	8.00	9.00
wine5	20	4.35	2.996	1	10	2.00	3.50	6.75
wine6	20	7.05	2.523	3	10	5.25	7.50	9.75
wine7	20	4.70	2.658	1	9	2.00	4.00	7.00
wine8	20	5.45	2.605	1	10	3.00	5.50	7.75
wine9	20	5.00	2.656	1	10	3.00	4.00	7.00
wine10	20	5.25	3.307	1	10	2.25	5.00	8.00

The second table outputted by SPSS is entitled 'Ranks'. At least numerically, the mean ranks were not equal. There was a mean rank as high as 7.25 (wine4) and as low as 4.35 (wine5). Whether the numerical differences amount to a statistically significant effect remains to be determined by the Friedman's ANOVA.

Ranks

	Mean Rank
wine1	5.00
wine2	5.90
wine3	5.05
wine4	7.25
wine5	4.35
wine6	7.05
wine7	4.70
wine8	5.45
wine9	5.00
wine10	5.25

Finally, SPSS outputted the Friedman's ANOVA result in the 'Test Statistics' table. It can be seen that the null hypothesis of equal mean ranks has been rejected, $\chi^2(9) = 18.23$, $p = .033$.

Test Statistics[a]

N	20
Chi-Square	18.229
df	9
Asymp. Sig.	.033

a. Friedman Test

Unfortunately, SPSS does not automatically report an estimate of effect size for Friedman's ANOVA. However, it can be calculated with the following partial η^2 formula:

$$\text{partial } \eta^2 = \frac{\chi^2_{Friedman}}{(N(k-1))} \qquad (5)$$

$$\text{partial } \eta^2 = \frac{18.23}{(20(10-1))} = \frac{18.23}{180} = .101$$

Thus, 10.1% of the variance in the ranks was accounted for by the type of wine, as rated by the 20 wine connoisseurs. In the context of a Friedman's ANOVA, the effect size is partial η^2, rather than eta-squared. Consequently, the percentage of variance accounted for in the ranks excludes the variance that is accounted for by the correlations across the conditions.

As it turns out, SPSS gives the menu option to estimate Kendall's *W*, which corresponds to partial η^2 in the Friedman's ANOVA case. When the analysis is run with the Kendall's *W* option selected, SPSS will include an additional row of results in the table entitled 'Test Statistics'. It can be seen that Kendall's *W* was estimated at .101, precisely the same estimate obtained from the partial η^2 formula used above. Thus, to save time, you might as well select the 'Kendall's W' option every time you perform a Friedman's ANOVA.

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Test Statistics

N	20
Kendall's W[a]	.101
Chi-Square	18.229
df	9
Asymp. Sig.	.033

a. Kendall's Coefficient of Concordance

In this fictional study, it was not necessary to conduct any follow-up multiple comparisons. It was only necessary to reject the omnibus null hypothesis, as such a result implies that the wine connoisseurs were able to agree, at least to some degree, on the quality rankings of the wines. If you were to follow-up with a multiple-comparison procedure, I would recommend a series of Friedman's ANOVAs testing the difference between two mean ranks, as the Wilcoxon-Signed-Rank test tests a slightly different hypothesis to that of the Friedman's ANOVA.

Friedman's ANOVA: Within-Subjects Mean Ranks – Ordinal Data

When the dependent variable is scored on a limited information ordinal scale (i.e., 4-point or 3-point scale), a non-parametric statistic should be considered, as the data have no hope of approximating a normal distribution. In the case of a within-subject design based on data associated with a limited information ordinal scale, a Friedman's ANOVA should be considered for the purposes of testing the null hypothesis of equal mean ranks, especially when there are three or more levels associated with the within-subjects factor.

Unfortunately, it is very difficult to find published studies that have measured a dependent variable on a 3- or 4-point scale in a within-subjects design with three or more levels. Consequently, for this demonstration, I simulated data for a fictional study.

Suppose a researcher wished to test the hypothesis that people's enjoyment of classical music increases as a function of exposure. To test the hypothesis, the researcher recruited 20 non-classical music listening participants aged 18 to 21 years of age. On four occasions, the participants listened to classical music for 90 minutes. On each occasion, the participants rated their experience associated with listening to classical music on a 4-point scale: 1 = horrible; 2 = unpleasant; 3 = pleasant; 4 = awesome. Because the scores were associated with only 4-points, the design is within-subjects, and there are more than two levels, the null hypothesis of equal mean ranks could be tested with Friedman's ANOVA (**Data File: classical_appreciation**). In contrast

Hypotheses

Null hypothesis (H_0): The mean ranks will be equal across all four conditions.

Alternative Hypothesis (H_1): The mean ranks will not be equal across all four conditions.

Friedman's ANOVA: Ordinal Data: SPSS

In order to conduct a Friedman's ANOVA in SPSS, one can use the 'Nonparametric Tests' menu option in SPSS (**Watch Video 16.12:** Friedman's ANOVA on Ordinal Data in SPSS). It is not necessary to transform the data into ranks. The Friedman ANOVA utility in SPSS will do so, automatically. As mentioned in the first example above, you are better off selecting Kendall's *W*, rather than Friedman's ANOVA, because the Kendall's *W* option will produce both the Friedman's ANOVA and the Kendall's *W* (partial η^2), simultaneously.

As can be seen in the SPSS output table entitled 'Ranks', the mean ranks increased numerically from the first listening session to the fourth listening session. The question is whether the numerical differences are statistically significant.

Ranks

	Mean Rank
first	1.73
second	2.53
third	2.83
fourth	2.93

As can be seen in the SPSS output table entitled 'Test Statistics', the Kendall's *W* (i.e., partial η^2) was estimated at .394. Furthermore, the corresponding Friedman's ANOVA analysis was statistically significant, $\chi^2(3) = 23.67$, $p < .001$. Thus, the null hypothesis of equal mean ranks was rejected. Furthermore, 39.4% of the variance in the ranks was accounted for by the conditions, excluding the variance accounted for by the correlations across the conditions.

Test Statistics

N	20
Kendall's W[a]	.394
Chi-Square	23.667
df	3
Asymp. Sig.	.000

a. Kendall's Coefficient of Concordance

Prior to carrying on with the analysis, I would like to point out the medians associated with the four conditions:

Statistics

		first	second	third	fourth
N	Valid	20	20	20	20
	Missing	60	60	60	60
Median		2.50	2.50	2.50	2.50

As can be seen in the SPSS table entitled ‘Statistics’, exposure to classical music did not change the level of appreciation even numerically, as each condition was associated with a median of 2.50. I believe this proves that the Friedman ANOVA is not a test of the difference between medians, as the Friedman ANOVA indicated a statistically significant effect. Again, Friedman’s ANOVA is a test of the difference between mean ranks, not medians (as many may lead you to believe). If you look at the raw data in the data file, the tendency was for some people to increase their appreciation from 1 to 2, and from 3 to 4. However, there were very few cases of changes from 2 to 3, which would have had an impact on the median. Although such a pattern of effects is unlikely to be observed in practice, it nonetheless underscores the fact that Friedman’s ANOVA has nothing to do with medians.

When applied to data with three or more levels, Friedman’s ANOVA is an omnibus test. Consequently, a statistically significant Friedman’s ANVOA simply suggests that at least one comparison amongst the mean ranks was statistically significant. It is necessary to conduct multiple comparison analyses to determine which of those pairwise comparisons were statistically significant, just as it is the case with a one-way within-subjects ANOVA.

Unfortunately, there are very few options to conducting multiple comparison tests in the context of analyses associated with within-subjects designs with data measured on a limited ordinal scale. Options include a series of Wilcoxon Signed-Rank Tests and the Sign Test.

Although rarely observed in practice, Friedman’s ANOVA can work perfectly well on data with only two levels, just like a regular ANOVA can be applied to interval/ratio data with only two levels. Zimmerman and Zumbo (1993) contended that Friedman’s ANOVA is a generalization of the less powerful sign test, however, I am not yet convinced that this is the case. Because the sign test, the Wilcoxon signed rank test, and Friedman’s ANOVA are all slightly different tests, I believe it is more logical to follow-up a statistically significant Friedman’s ANOVA with a series of Friedman’s ANVOA multiple comparisons. That is, in such a scenario, the results are directly comparable. Additionally, in SPSS, when a Friedman’s ANOVA is performed, the option to estimate simultaneously an estimate of effect size (partial eta-squared via Kendall’s *W*) is available.

Friedman’s ANOVA: Ordinal Data: SPSS – Post-Hoc Testing

To perform a series of multiple comparison Friedman’s ANOVAs, simply perform the Friedman’s ANOVA as described above. However, include only two conditions in each analysis. In this case, with four conditions, there are a total of six possible pairwise comparisons. For the purposes of comparison, I have included the results associated with the sign-test (binomial *p*-value), the Wilcoxon Signed-Rank Test (I squared the *Z*-value to yield a chi-square value), and Friedman's ANOVA. As can be seen in the Table C16.3, the sign-test yielded the largest *p*-values, which suggests that it was the most conservative test. By contrast, the Wilcoxon Signed-Rank Test (W-S-R-T) and the Friedman's ANOVA (F-A) yielded nearly identical results.

Table C16.3. Results Associated with the Multiple-Comparison Procedures

			Sign		W S-R T		Friedman	
Comparison			Sign	p	χ^2	p	χ^2	p
First	vs.	Second	--	.008	8.00	.005	8.00	.005
First	vs.	Third	--	.001	11.00	.001	11.00	.001
First	vs.	Fourth	--	.002	9.94	.002	10.29	.001
Second	vs.	Third	--	.250	3.00	.083	3.00	.083
Second	vs.	Fourth	--	.219	2.78	.096	2.67	.102
Third	vs.	Fourth	--	1.00	.67	.414	.33	.564

Note. Sign = the sign-test is a binomial test, consequently, the analysis yields only a *p*-value; W S-R T = Wilcoxon Signed-Rank-Test; Friedman = Friedman's ANOVA.

Technically, it would be appropriate to apply some sort of the adjustment to the *p*-values to ensure that the familywise error rate does not exceed .05. One could apply a Bonferroni adjustment by multiplying the multiple-comparison *p*-values by the number of comparisons. In this case, the number of comparisons performed was equal to 6. Thus, the Friedman's ANVOA adjusted *p*-values are: .030, .006, .006, .498, .612, and .999 (theoretically, .9999 is the highest level of probability). Thus, the exposure to classical music increased appreciation, however, the effect was only significant for comparisons that included the first listening experience.

Friedman's ANOVA: Very Skewed Data

By far, the most common use of Friedman's ANOVA is where the data are determined to be insufficiently normally distributed for the purposes of performing a parametric statistic. As I have argued in other chapters, parametric statistics such as the one-way within-subjects ANOVA, for example, tend to be relatively robust to violations of normality. Specifically, based on simulation research, the *t*-test and the ANOVA tend to be robust when skew is < |2.0| and kurtosis is < |9.0|. Consequently, it should only be in a small minority of cases that a researcher should feel compelled to use a statistic such as Friedman's ANOVA, because the interval/ratio data are not sufficiently normally distributed for the purposes of conducting a parametric statistical analysis.

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In some cases, however, the data will be severely non-normally distributed, and the Friedman's ANOVA should be considered for testing the hypothesis of unequal levels of central tendency, even if the inference is restricted to equal mean ranks. Consider, for example, Lundrigan and Canter (2001) who were interested in determining whether serial murders travel progressively further distances (in kilometers) to dispose of the bodies. To examine the question, Lundrigan and Canter (2001) collected archival information from the police files associated with 71 American serial murders who murdered at least 6 individuals. I have simulated data to replicate fairly closely the results reported by Lundrigan and Canter (2001) (**Data File: serial_murder_disposal**).

First, I will note that the data were associated with substantial skew. Specifically, as can be seen in the SPSS table entitled 'Statistics', the skew associated with the first body disposal distribution was in excess of |2.0|. Consequently, even though the dependent variable was measured on a ratio scale, it would be arguably inappropriate to apply a parametric statistical analysis (e.g., one-way within-subjects ANOVA) in this case. Instead, a Friedman's ANOVA would be more appropriate, as the analysis does not assume any level of normality (**Watch Video 16.13:** Friedman's on Skewed Data in SPSS).

Statistics

		body1	body2	body3	body4	body5	body6
N	Valid	71	71	71	71	71	71
	Missing	0	0	0	0	0	0
Mean		31.79	33.83	40.27	44.97	50.03	61.80
Median		21.00	22.29	29.53	32.06	36.22	60.66
Std. Deviation		36.86	36.42	41.04	41.57	41.83	39.75
Skewness		2.11	1.46	1.60	1.51	1.33	.81
Kurtosis		4.38	1.41	2.94	2.33	1.96	1.62
Minimum		.00	.00	.00	.00	.00	.00
Maximum		173.00	138.15	198.31	200.30	195.00	208.00

As can be seen in the SPSS table entitled 'Ranks', the mean ranks ranged from 2.88 to 4.55.

Ranks

	Mean Rank
body1	2.88
body2	2.97
body3	3.11
body4	3.52
body5	3.96
body6	4.55

Furthermore, as can be seen in the SPSS table entitled 'Test Statistics', the null hypothesis of equal mean ranks was rejected, $\chi^2(5) = 43.43$, $p < .001$.[2] Finally, Kendall's *W* was estimated at .122, which implies that 12.2% of the variance in the kilometers travelled (ranked) was accounted for by the number of murders committed. Given the direction of the ranks, it may be suggested that the offenders travelled further and further to dispose of the bodies, as the number of offences they committed increased.

Test Statistics

N	71
Kendall's W[a]	.122
Chi-Square	43.429
df	5
Asymp. Sig.	.000

a. Kendall's Coefficient of Concordance

At this point, many researchers would follow-up the statistically significant omnibus effect with a series of multiple-comparisons to uncover precisely which of the six levels in the independent variable were statistically significantly different from each other. In fact, Lundrigan and Caner (2001) did just that and reported seven comparisons as statistically significant, based on a series of Wilcoxon signed-rank tests. However, Lundrigan and Cantor (2001) did not make any adjustments to help ensure the familywise error rate remained close to .05. Based on my calculations, none of the comparisons would have been significant statistically after Bonferroni adjustment. To learn about a much more powerful approach, I describe a trend analysis based on these data in the Advanced Topics section of this chapter.

Cochran's Q

Anecdotal evidence suggests that university students tend not to be naturally interested in taking a research methods or statistics unit. However, for many degrees, students must complete at least one researcher methods or statistics unit. To examine the issue more scientifically, I once surveyed my students with the following question at the beginning of the semester: "I'm taking this unit simply because I have to'. Approximately 82% of the students agreed with that statement. I had my work cut out for me!

When I am at my most self-indulgent, I like to think that the quality of my teaching effects a statistically significant reduction in the percentage of students who endorse the above item across the semester. Ideally, I would run a study where I asked the above item in

[2] Lundrigan and Canter (2001) reported a chi-square value of 14.95, however, from what I can tell, such a value was based on an analysis that included the sample of serial killers who committed up to 10 murders.

the first week of the semester, in the middle of the semester, and, finally, at the end of the semester. If I asked the above question with a nominally scaled response scale (yes/no), any numerical changes in the percentage of the students who endorse the item could be tested for statistical significance with Cochran's Q. McNemar's chi-square would not apply (Chapter 4), because it can only handle two levels. My proposed study has three levels: beginning, middle, and end (**Data File: students_hate_statistics**).

Hypotheses

Null Hypothesis (H_0): The within-subject percentages will be equal across the semester.

Alternative Hypothesis (H_1): The within-subject percentages will be unequal across the semester.

Cochran's Q: SPSS

In order to conduct a Cochran's Q analysis in SPSS, you can use the 'Nonparametrics' menu option in SPSS (**Watch Video 16.14:** Cochran's Q in SPSS). Prior to consulting the actual Cochran's Q analysis results, it is best to examine the descriptive statistics. As can be seen in the SPSS table entitled 'Descriptive Statistics', the means associated with each of the three time periods corresponded to .846, .750, and .635. When a mean is calculated from a variable that is scored dichotomously 0 and 1, it represents the proportion. Expressed as percentages, the percentage of students who responded that they were only taking the unit 'because they had to' was equal to 84.6%. However, by mid-semester, the percentage dropped (at least numerically) to 75.0%. Finally, by the end of the semester, the percentage was equal to 63.5%. The statistical question is whether these percentages are different from each other in a statistically significant way or not.

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum
semester_start	52	.846	.364	0	1
semster_mid	52	.750	.437	0	1
semester_end	52	.635	.486	0	1

As can be seen in the SPSS table entitled 'Test Statistics', the null hypothesis of equal percentages was rejected, $\chi^2 = 16.55$, $p < .001$. However, because the analysis included three percentages, it is not known which percentages were different from each other. That is, Cochran's Q applied to three or more percentages is an omnibus statistic. A statistically significant effect needs to be followed-up with multiple comparisons. Unfortunately, here are no dedicated multiple comparisons procedures for comparisons between within-subjects

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percentages. Consequently, a series of three Cochran's Q analyses could be performed. In this case, as there were only three levels within the factor (beginning, middle, and end of semester), I believe the follow-up analyses would be protected, based on the same logic and research which supports the protected Fisher's LSD (see Chapter 6).

Test Statistics

N	52
Cochran's Q	16.545[a]
df	2
Asymp. Sig.	.000

a. 0 is treated as a success.

As can be seen in Table C16.4, the start versus end-of-semester comparison yielded a statistically significant effect, χ^2 = 11.00, p < .001 Thus, the percentage of students who responded 'yes' to the "…because I have to" item decreased from the start of the semester to the end, χ^2 = 5.00, p = .025. Furthermore, there was a statistically significant decrease from start to mid, and a statistically significant decrease from mid-semester to the end of the semester, χ^2 = 6.00, p = .014.

Table C16.4: Cochran's Q Multiple Comparison Results

Start vs. End

Test Statistics

N	52
Cochran's Q	11.000[a]
df	1
Asymp. Sig.	.001

a. 0 is treated as a success.

Start vs. Mid

Test Statistics

N	52
Cochran's Q	5.000[a]
df	1
Asymp. Sig.	.025

a. 0 is treated as a success.

Mid. Vs. End

Test Statistics

N	52
Cochran's Q	6.000[a]
df	1
Asymp. Sig.	.014

a. 0 is treated as a success.

Non-Parametrics: Summary

I believe non-parametric analyses have a place in the data analyst's tool belt. However, as a general statement, I would go to great lengths to avoid using them. My point of view is not based on the commonly articulated notion that non-parametric analyses tend to be less powerful than parametric statistics. Such a position is not based on the facts: there are clear cases where a non-parametric analysis will be more powerful than a corresponding parametric analysis (Zimmerman & Zumbo, 1993a). Instead, my point of view is based on my strong preference to keep as close as possible to the raw data, because the inferences made from a statistical analysis will be based on the nature of the data that were analyzed, not the data that were collected. As a researcher, I don't want to make inferences about mean ranks, I want to make inferences about mean kilometers, IQ scores, or reaction time, for example.

Advanced Topics

A number of topics covered in this section of the chapter include the application of well-known parametric statistics on ranked data. The research in the area suggests that tests such as the *t*-test and the ANOVA *F*-test yield accurate *p*-values, when they are applied to the test of the difference between mean ranks (Conover & Iman, 1976; Zimmerman & Zumbo, 1993a; Zimmerman & Zumbo, 1993b; Zimmerman & Zumbo, 1993c). I'm big fan of this approach, as it demystifies the nature of the non-parametric statistics – they are not that different from parametric statistics. Additionally, parametric statistics applied on ranks opens-up the opportunity to conduct very useful analyses such as the Welch's *t*-test and the Brown-Forsythe *F*-test, neither of which assumes homogeneity of variance or equal sample sizes.

Independent Samples *t*-test on the Ranks: Example 1

Someone who knows a little about statistics may think that nonparametric statistics are qualitatively different to parametric statistics. However, such an impression is mistaken. For example, it is perfectly acceptable to convert data into ranks and then perform an independent samples *t*-test on those ranks (Conover & Iman, 1976; Zimmerman & Zumbo, 1993c). The results obtained from the *t*-test on ranks will be accurate, potentially even more accurate than the conventional corresponding nonparametric statistic (e.g., Mann-Whitney)! Because parametric statistics can be performed on ranks, the number of options to conduct various multiple comparisons, as well as analyses within factorial designs with ordinal/ranked data, increases, as there are very few options within a non-parametric framework. Similarly, the opportunity to employ Welch's *t*-test, the Brown-Forsythe *F*-test is made available, thus, the problem of unequal variances and unequal samples sizes vanishes. If the application of a parametric statistic to ranked data feels odd to you, consider that the Spearman (rank) correlation is based on the very same principal: the data are ranked and a parametric statistic is applied to those ranked data (i.e., Pearson's *r*) to estimate the association between two variables.

In order to conduct an independent samples *t*-test on ranks (Conover & Inman, 1976), simply convert the dependent variable scores into ranks, and then conduct a regular independent samples *t*-test on the ranks. Data can be converted into ranks in SPSS by using the transform utility. Then conduct a regular independent samples *t*-test on the ranked data (**Data File: MEQ_data**). (**Watch Video 16.15:** Independent t-test on Ranks in SPSS).

As can be seen in the SPSS table entitled 'Group Statistics', the males and females were associated with a mean rank of 954.50 and 1176.12, respectively. The standard deviations were also reported.

Group Statistics

	gender	N	Mean	Std. Deviation	Std. Error Mean
Rank of MEQ16	Male	1041	954.50432	572.018473	17.729018
	Female	1094	1176.12066	573.488237	17.338673

In the next SPSS table entitled 'Independent Samples Test', SPSS reported the results associated with the independent samples *t*-test on ranks.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
Rank of MEQ16	Equal variances assumed	.503	.478	-8.936	2133	.000	-221.61634	24.799719	-270.250	-172.982
	Equal variances not assumed			-8.937	2128.27	.000	-221.61634	24.798138	-270.247	-172.985

As can be seen in the SPSS table entitled 'Independent Samples Test', the *t*-test on the ranks was estimated at t = -8.936 (p < .001), which is very comparable to the Z-value obtained from the Mann-Whitney (-8.78). The fact that the Z and t-values were very similar is not a coincidence. A Mann-Whitney test and an independent samples *t*-test on ranks are essentially the same test (Conover & Inman, 1976).

In order to estimate η^2 in the independent samples *t*-test case, the following formula can be used:

$$\eta^2 = \frac{t^2}{t^2 * df_{\text{Within}}} \tag{6}$$

$$\eta^2 = \frac{-8.94^2}{-8.94^2 * 2133} \frac{79.92}{79.92 + 2133} = .0361$$

Thus, 3.61% of the variance in the ranks was accounted for by sex.

Because the standard deviations were reported in the independent samples *t*-test on the ranks case, Cohen's *d* can also be estimated based on the conventional formula:

$$d = \frac{M_1 - M_2}{(SD_1 + SD_2)/2} \tag{7}$$

$$d = \frac{954.50 - 1176.12}{(573.49 + 572.02)/2} = \frac{-221.62}{572.76} = -.387$$

A Cohen's *d* value of -.39 corresponds to somewhere between a small and medium effect size (Cohen, 1988; 1992). The clear conclusions that should be drawn from the *t*-test on ranks results is that he Mann-Whitney test is not some magical that does not have assumptions. Instead, it is fundamentally similar to the independent samples *t*-test on ranks. Secondly, the opportunity to perform parametric statistics on ranked data opens up a lot of possibilities, as discussed further below.

Independent Samples *t*-test on the Ranks: Example 2

Personally, I would prefer not to report the mean ranks that SPSS reported in this Mann-Whitney case (see in Foundations section of this chapter: 'Mann-Whitney on Original Ranks: Example 2'), because they have no intuitive interpretation (i.e., 87.65 vs. 185.35). However, if I tested the null hypothesis by performing an independent samples *t*-test on ranks, I would obtain the intuitive mean ranks, as well as an identical statistical result.

Group Statistics

	sex	N	Mean	Std. Deviation	Std. Error Mean
earning_capacity	Males	96	4.26	2.502	.255
	Females	213	7.42	2.992	.205

As can be seen in the SPSS table entitled 'Group Statistics' earning capacity was associated with mean ranks of 4.26 and 7.42 for the males and females, respectively. Thus, females ranked earning capacity as more attractive than males, at least numerically (a higher rank value = more desirable).

I tested the assumption of homogeneity of variances (via the Explore utility). As can be seen in the table entitled 'Test of Homogeneity of Variance', the homogeneity of variances assumption was violated based on the median version of the test (adjusted df), $F(1, 202.66) = 3.98$, $p = .047$. Thus, the Mann-Whitney test should not be applied in this case, particularly considering the sample sizes are very unequal (96 vs. 213). Instead, the Welch's *t*-test on ranks can be employed, as it does not assume homogeneity of variances or equal sample sizes. The opportunity to employ Welch's *t*-test is a great advantage of applying a parametric statistic on ranked data.

Test of Homogeneity of Variance

		Levene Statistic	df1	df2	Sig.
earning_capacity	Based on Mean	5.398	1	307	.021
	Based on Median	3.975	1	307	.047
	Based on Median and with adjusted df	3.975	1	303.659	.047
	Based on trimmed mean	6.675	1	307	.010

As can be seen in the SPSS table entitled 'Independent Samples Test', the difference in the mean ranks was significant statistically, $t(216.65) = -9.66$, $p < .001$, based on the Welch's *t*-

test (i.e., equal variances not assumed). Cohen's *d* was estimated at d = -1.15 (4.26 – 7.42) / ((2.50 + 2.99) / 2) which is a large effect size, based on Cohen's (1988; 1992) guidelines. Thus, females rate earning capacity as a much more attractive quality in a potential mate, in comparison to males.

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means		
		F	Sig.	t	df	Sig. (2-tailed)
earning_capacity	Equal variances assumed	5.398	.021	-9.028	307	.000
	Equal variances not assumed			-9.657	216.646	.000

One-Way Between-Subjects ANOVA on Ranks

Some may be surprised that it is possible to calculate η^2 from a non-parametric statistic such as the Kruskal-Wallis (see formula 3 in this chapter). However, this impression is based on the misperception that there is a meaningful difference between the Kruskal-Wallis test and the one-way between subjects ANOVA. There is essentially no difference between the two tests, when the one-way between-subjects ANOVA is applied on dependent variable data that has been transformed into ranks (Conover & Iman, 1976; Zimmerman & Zumbo, 1993c).

To illustrate, I converted the ordinal response scale data associated with the Cook et al. (2011) dependent variable data into ranks (**Data File: veggies_oneway_between_ranks**). I then performed a regular one-way between-subjects ANVOA on the ranks. As can be seen in the SPSS table entitled 'ANOVA', the null hypothesis of equal mean ranks was rejected, F(3, 418) = 19.69, p < .001. Furthermore, the effect size was estimated at η^2 .124 (553363.763 / 4468860.00 = .124), which was identical to the result obtained from the Kruskal-Wallis (**Watch Video 16.16:** Between-Subjects ANOVA on Ranks in SPSS).

ANOVA

rated_liking_ranks

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	553363.763	3	184454.588	19.692	.000
Within Groups	3915496.24	418	9367.216		
Total	4468860.00	421			

The fact that ANOVA on the ranks was essentially identical to the Kruskal-Wallis opens up the possibility of overcoming a serious limitation with non-parametric statistics such as Kruskal-Wallis: the impossibility of testing factorial design hypotheses with non-parametric statistics (most software packages do not offer such opportunities). However, given the plausibility of ANOVA on the ranks, the real possibility of testing factorial design (e.g., 2x2 between-subjects factorial design; 2x2 mixed design ANOVA) hypotheses with ordinal dependent variables is made available, as I discuss further below. Additionally, there are many more parametric multiple comparison procedures that can be employed on ranked data, in comparison to non-parametric dedicated multiple comparison procedures.

Brown-Forsythe *F*-Test: Unequal Variances

The one-way between-subjects ANOVA on ranks is an attractive option. However, when the data do not satisfy the assumption of equal distribution shapes (i.e., as tested via Levene's test of homogeneity of variance), and the sample sizes are unequal, an alternative analysis must be sought. As described in more detail in the one-way between-subjects chapter, the Brown-Forsythe *F*-test is an excellent alternative to the one-way between-subjects ANOVA, when the variances are unequal and the sample sizes are unequal. Consequently, I revisited the Cook et al. (2011) study (**Data File: veggies_oneway_between_ranks**) discussed in the Foundations section of this chapter relevant to rewarding children for eating vegetables. You may recall that the assumption of homogeneity of variance was violated, $F(3, 402.03) = 19.88$, $p < .001$. Consequently, I conducted a Brown-Forsythe *F*-test on the ranks in SPSS (**Watch Video 16.17:** Brown-Forsythe on Ranks in SPSS). As can be seen in the SPSS table entitled 'Robust Tests of Equality of Means', the null hypothesis of equal means has been rejected, $F(3, 402.03) = 19.88$, $p < .001$.

Robust Tests of Equality of Means

rated_liking_ranks

	Statistic[a]	df1	df2	Sig.
Brown-Forsythe	19.876	3	402.031	.000

a. Asymptotically F distributed.

Thus, the null hypothesis of equal mean ranks has been rejected, and I'm not concerned that the homogeneity of distributional shapes violation has been violated, as the Brown-Forsythe test should offer some protection, in relation to that issue. Like any other ANOVA with more than two groups, I only know that all of the mean ranks are not equal. I do not know which specific comparisons (of only two means) are statistically significant. Although I could apply a series of Brown-Forsythe multiple comparisons, there are more than three groups in this study (i.e., 4), therefore, I would have to consider familywise inflation and perhaps implement a Bonferroni correction on the p-values. A more attractive option, in my view, would be to apply the Games-Howell multiple comparison procedure on these data, which I do next.

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Robust Non-Parametric Between-Subjects Multiple Comparison: Games-Howell

An excellent multiple comparison option is the Games-Howell multiple comparison procedure. The Games-Howell multiple comparison procedure does not assume homogeneity of variance or equal sample sizes (Dunnett, 1980). Furthermore, the *p*-values do not need to be adjusted via Bonferroni, as it is a specifically designed multiple comparison procedure that keeps familywise error at .05, in most cases. Although the Games-Howell procedure simulation research has been conducted on interval/ratio data, I am very confident that the strengths associated with the Games-Howell procedure extend to applications on ranks, based on the simulation research conducted by Zimmerman and Zumbo (1993c). Additionally, if the data were severely non-normally distributed, the Games-Howell procedure could be estimated via bootstrapping, an estimation technique that does not assume any level of measurement or any level of normality.

In the Cook et al. (2001) study (see Foundations section of this chapter), the dependent variable was measured on a limited ordinal scale (3-point Likert) across the four groups. Furthermore, the homogeneity of variance assumption was violated and the sample sizes were unequal. Arguably, an attractive approach to testing the null hypothesis of equal mean ranks is the bootstrapped Games-Howell (**Watch Video 16.18:** Games-Howell on Limited Ordinal in SPSS), which is a single-step multiple comparison procedure. When I tested the hypothesis of equal means across the four groups, I obtained the results reported in the SPSS table entitled, 'Bootstrap for Multiple Comparisons'. It can be seen that control group had a statistically significantly lower mean liking rating than the three treatment groups, as the lower and upper confidence intervals did not intersect with zero. Such a result is similar to that found with these data when analysed with the Kruskal-Wallis/Mann-Whitney post-hoc tests. However, a key difference is that the Exposure + Praise group reported a statistically significantly larger mean liking rating (mean difference = .193) than the Exposure Only group (95%CI: .033/.348). Interestingly, the more conventional Mann-Whitney U tests with Bonferroni correction reported in the Foundations section of this chapter failed to find this mean difference to be significant ($p = .078$). So, the Games-Howell bootstrapped procedure on the raw data could prove to be a more powerful approach to analyze your data! Also, a major upshot is that you don't have to talk about mean ranks, as per conventional non-parametric statistics. Instead, you can talk the results on the basis that they were actually measured.

Bootstrap for Multiple Comparisons

Dependent Variable: rated_liking
Games-Howell

(I) group	(J) group	Mean Difference (I-J)	Bootstrap[a] Bias	Bootstrap[a] Std. Error	BCa 95% Confidence Interval Lower	BCa 95% Confidence Interval Upper
Control	Exposure + Tangible	-.590	-.004	.105	-.788	-.389
	Exposure + Praise	-.638	.000	.097	-.825	-.439
	Exposure Only	-.445	.000	.104	-.654	-.234
Exposure + Tangible	Control	.590	.004	.105	.372	.810
	Exposure + Praise	-.048	.004	.084	-.222	.121
	Exposure Only	.145	.004	.093	-.055	.330
Exposure + Praise	Control	.638	.000	.097	.441	.823
	Exposure + Tangible	.048	-.004	.084	-.110	.204
	Exposure Only	.193	.000	.079	.033	.348
Exposure Only	Control	.445	.000	.104	.235	.652
	Exposure + Tangible	-.145	-.004	.093	-.314	.013
	Exposure + Praise	-.193	.000	.079	-.339	-.043

a. Unless otherwise noted, bootstrap results are based on 1000 bootstrap samples

For comparison sake, I also performed the Games-Howell procedure on the rank transformation of the dependent variable (**Data File: veggies_oneway_between_ranks**). As can be seen in the SPSS table entitled 'Multiple Comparisons', all of the comparisons with the control condition were statistically significant. It will be noted that several of the other comparisons were nearly significant. For example, the 'exposure + praise' mean rank versus the 'exposure only' mean rank comparison was associated with a *p*-value of .060. When performed via multiple Mann-Whitney tests with the Bonferroni adjustment, the corresponding *p*-value was estimated at .078. It's not a massive difference, but why throw away precious power, when you don't have to?

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Multiple Comparisons

Dependent Variable: rated_liking_ranks
Games-Howell

(I) group	(J) group	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
Control	Exposure + Tangible	-88.06*	14.07	.000	-124.51	-51.61
	Exposure + Praise	-86.78*	13.30	.000	-121.22	-52.33
	Exposure Only	-55.66*	14.28	.001	-92.62	-18.69
Exposure + Tangible	Control	88.06*	14.07	.000	51.61	124.51
	Exposure + Praise	1.29	12.13	1.00	-30.14	32.72
	Exposure Only	32.41	13.20	.070	-1.78	66.59
Exposure + Praise	Control	86.78*	13.30	.000	52.33	121.22
	Exposure + Tangible	-1.29	12.13	1.00	-32.72	30.14
	Exposure Only	31.12	12.37	.060	-.91	63.15
Exposure Only	Control	55.66*	14.28	.001	18.69	92.62
	Exposure + Tangible	-32.41	13.20	.070	-66.59	1.78
	Exposure + Praise	-31.12	12.37	.060	-63.15	.91

*. The mean difference is significant at the 0.05 level.

Kruskal-Wallis Between-Subjects Factorial Design

There are no well-known or well-established nonparametric statistics that can be applied to factorial designs with dependent variables measured on an ordinal (ranked) scale. Furthermore, to my knowledge, there are no well-known statistics packages that allow users to test factorial design interactions with ordinal data. However, given the close similarity between tests such as Kruskal-Wallis and ANOVA on ranks, it was demonstrated that a one-way between-subjects ANOVA on ranks can be performed, justifiably. Factorial design analyses on ranks are more complicated. Specifically, it has been discovered that testing interactions on ranked data results in unpredictable results. Because ranking data is a non-linear transformation, main effects and/or interactions can variously appear or disappear, when a factorial ANOVA has been performed on ranked data (Higgins & Tashtoush, 1994). In order to correct the problem, Higgins and Tashtoush (1994) introduced a modification to the ANOVA on ranks procedure. The modification is known as 'aligned rank transform'. I hope to demonstrate this procedure in a future edition of this text.

A Note on Friedman's ANOVA with Two Levels

Almost invariably, researchers use Friedman's ANOVA when there are three or more levels in the independent variable. However, Friedman's ANOVA works just fine when there are only two levels in the independent variable. Furthermore, in some cases, it can be a more powerful test than the Wilcoxon Signed-Rank test. Finally, I prefer Friedman's, because it is relatively similar to the oneway within-subjects ANOVA, which allows for the estimation of a

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more clearly interpretable partial η^2 (as described below). Consequently, when there are only two levels in the within-subjects factor, you should seriously consider application of the Friedman's ANOVA in the non-parametric within-subjects design with two levels case.

In light of the above, I re-visited the junior and senior doctor study relevant to confidence described in the Foundations section of this chapter (**Data File: doctors_confidence**). Specifically, I tested the null hypothesis of equal mean ranks between the junior doctors (confidence ratings) and the senior doctors (competence ratings) with Friedman's ANOVA. As can be seen in the table entitled 'Ranks', the junior doctors confidence ratings were associated with a mean rank of 1.92. By contrast, the senior doctor competence ratings were associated with a mean competence rating of 1.08. Thus, junior doctors rated themselves as numerically more confident than the senior doctor ratings of competence.

Ranks

	Mean Rank
confidence	1.92
competence	1.08

Next, as can be seen in the SPSS table entitled 'Test Statistics', the null hypothesis of equal mean ranks has been rejected, $\chi^2(1) = 25.00$, $p < .001$. Thus, the numerical difference between the mean ranks was statistically significant. For the purposes of comparison, I'll note that the Friedman ANOVA p-value to seven decimal places was .0000006. By comparison, the Wilcoxon Sign-Rank Test p-value reported in the Foundations section of this chapter was actually .0000007. It is not a large difference between the two p-values, however, there may be cases where the Wilcoxon Sign-Ranked Test will yield a p-value of .055 and Friedman's ANOVA will yield a p-value of .045 (**Watch Video 16.19:** Friedman's ANOVA for Two Mean Ranks in SPSS).

Test Statistics[a]

N	30
Chi-Square	25.000
df	1
Asymp. Sig.	.000

a. Friedman Test

In addition to the possibility of greater statistical power, the Friedman's ANOVA offers the opportunity to estimate a fully interpretable partial η^2 estimate with the following formula:

$$\text{partial}\,\eta^2 = \frac{\chi^2}{(N(k-1)} \qquad (8)$$

Where χ^2 = the χ^2-value test statistic; N = total number of pairings; and k = number of levels.

$$\text{partial}\,\eta^2 = \frac{25.00}{30} = .833$$

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Thus, 83.3% of the variance in the ranks was accounted for by the independent variable, controlling for the shared variance between the confidence and competence ratings.

If you were really ambitious, you could convert the ratings in the data file into ranks and then perform a paired samples-test or a repeated measures ANOVA on the ranked data. Either analysis would test the null hypothesis of equal mean ranks. Furthermore, the *p*-value and the effect size estimates would be accurate (Zimmerman & Zumbo, 1993b). In this case, the one-way within-subjects ANOVA on ranks rejected the null hypothesis and estimated the partial η^2 at .833, as can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects'. It is not a coincidence that the Friedman ANOVA and the one-way within-subjects ANOVA on ranks estimated the same partial η^2 (**Watch Video 16.20:** [Within Subjects ANOVA on Ranks (2 Levels) in SPSS]).

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Sphericity Assumed	10.417	1	10.417	145.000	.000	.833
	Greenhouse-Geisser	10.417	1.000	10.417	145.000	.000	.833
	Huynh-Feldt	10.417	1.000	10.417	145.000	.000	.833
	Lower-bound	10.417	1.000	10.417	145.000	.000	.833
Error(factor1)	Sphericity Assumed	2.083	29	.072			
	Greenhouse-Geisser	2.083	29.000	.072			
	Huynh-Feldt	2.083	29.000	.072			
	Lower-bound	2.083	29.000	.072			

For thoroughness, I have also conducted a paired samples *t*-test on ranks, which rejected the null hypothesis of equal mean ranks, $t(29) = 12.04$, $p < .001$. A *t*-value is equal to an *F*-value when the *t*-value is squared. Thus, $12.042^2 = 145.009$, which is essentially identical to the one-way within-subjects ANOVA result. Although the one-way within-subjects ANOVA and paired samples *t*-test on ranks are attractive options in the case of the testing the null hypothesis of equal within-subjects mean ranks, my inclination would be to employ the Friedman's ANOVA, if I were writing a manuscript, if only because I couldn't be bothered trying to explain to the reviewers/readers that it is a perfectly justifiable procedure.

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference: Lower	Upper			
Pair 1	confidence_rank - competence_rank	.833	.379	.069	.692	.975	12.042	29	.000

One-Way Within-Subjects ANOVA on Ranks

Although not well-known, it is perfectly acceptable to perform a one-way within-subject ANOVA on ranked data. The *p*-value obtained from such an analysis will keep the Type I error close to .05 (Zimmerman & Zumbo, 1993b). Additionally, the one-way within subjects ANOVA on ranks has been found to be somewhat more powerful than Friedman's ANOVA in many cases, particularly when the data approximate a normal, uniform, or exponential distribution (Zimmerman & Zumbo, 1993b). For example, Zimmerman and Zumbo (1993b) found that, with four repeated measurement levels (N = 14), the one-way within-subjects ANOVA on ranks was associated with a power of .901, whereas Friedman's ANOVA was associated with power of .823 (various levels of effect size were specified; normal distribution). Friedman's ANOVA is preferred for data consistent with the Cauchy distribution, however.

One-Way Within-Subjects ANOVA on Ranks: SPSS (Example 1)

In the wine tasting study I described in the Foundations section of this chapter (**Data File: wine_tasting**), the data were ranked to begin with. Consequently, it was not necessary to rank the data prior to conducting the analysis. A one-way within-subject ANOVA can be conducted in the typical fashion (see chapter on one-way within-subjects ANOVA; such knowledge is assumed here), as the only difference is that the data are based on ranks. As can be seen in the SPSS table entitled 'Mauchly's Test of Sphericity', the one-way within-subjects ANOVA assumption of sphericity has been satisfied, $\chi^2(44) = 52.47$, $p = .205$, which I believe will always be the case with ranked data (**Watch Video 16.21:** Within-Subjects ANOVA on Ranks (2 or More Levels) in SPSS).

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Huynh-Feldt	Lower-bound
factor1	.036	52.470	44	.205	.639	.947	.111

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: factor1

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Next, as can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects', the null hypothesis of equal mean ranks has been rejected, $F(9, 171) = 2.14$, $p = .029$, partial η^2 = .101. Notice how the partial η^2 estimated here with the one-way within-subjects ANOVA is identical to the partial η^2 estimated from Kendall's *W* (see Foundations section of this chapter). Thus, I hope this example inspires you with confidence that the oneway within-subjects ANOVA can be applied justifiably and insightfully to ranked data (Zimmerman & Zumbo, 1993b).

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Sphericity Assumed	167.100	9	18.567	2.141	.029	.101
	Greenhouse-Geisser	167.100	5.749	29.066	2.141	.057	.101
	Huynh-Feldt	167.100	8.526	19.599	2.141	.032	.101
	Lower-bound	167.100	1.000	167.100	2.141	.160	.101
Error(factor1)	Sphericity Assumed	1482.900	171	8.672			
	Greenhouse-Geisser	1482.900	109.230	13.576			
	Huynh-Feldt	1482.900	161.994	9.154			
	Lower-bound	1482.900	19.000	78.047			

One-Way Within-Subjects ANOVA on Ranks – SPSS (Example 2)

In this section of the Advanced Topics section of the chapter, I re-visit the classical music exposure example described in the Foundations section of this chapter (**Data File: classical_appreciation**). However, instead of Friedman's ANOVA, I will re-test the null hypothesis by performing a regular within-subjects ANOVA on ranks. I had to rank the data myself. Because the ranking process in the Friedman's ANOVA is within a subject, not across subjects, it is a bit more work to obtain the ranks in SPSS. After the four modified variables (ranked) variables have been created (i.e., there are four levels in the within-subjects factor), a regular one-way within-subjects ANVOA can be performed on the ranked data (**Watch Video 16.22: Within-Subjects ANOVA on Ranks 3**).

As can be seen in the SPSS table entitled 'Descriptive Statistics', the mean ranks wre identical to those obtained from the Friedman's ANOVA. Additionally, the standard deviations were reported, which is valuable extra information that was not reported in the Friedman's ANOVA output.

Descriptive Statistics

	Mean	Std. Deviation	N
first_rank	1.725	.7159	20
second_rank	2.525	.5495	20
third_rank	2.825	.4375	20
fourth_rank	2.925	.6544	20

Next, the assumption of sphericity was tested and satisfied, $\chi^2(5) = 7.20$ $p = .207$, as can be seen in the SPSS table entitled 'Mauchly's Test of Sphericity'.

Mauchly's Test of Sphericity[a]

Measure: MEASURE_1

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon[b] Greenhouse-Geisser	Huynh-Feldt	Lower-bound
factor1	.666	7.201	5	.207	.818	.949	.333

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

As can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects', the null hypothesis of equal mean ranks was tested and rejected, $F(3, 57) = 12.38$, $p < .001$, partial η^2 = .394. Notice how the partial η^2 is identical to the Kendall's W estimate yielded in the Friedman's ANOVA analysis: they represent the same statistical effect. Thus, 39.4% of the variance in the ranks was accounted for by the independent variable (exposure to classical music), controlling for the correlation between the conditions.

Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Sphericity Assumed	17.750	3	5.917	12.376	.000	.394
	Greenhouse-Geisser	17.750	2.454	7.232	12.376	.000	.394
	Huynh-Feldt	17.750	2.846	6.237	12.376	.000	.394
	Lower-bound	17.750	1.000	17.750	12.376	.002	.394
Error (factor1)	Sphericity Assumed	27.250	57	.478			
	Greenhouse-Geisser	27.250	46.631	.584			
	Huynh-Feldt	27.250	54.069	.504			
	Lower-bound	27.250	19.000	1.434			

Finally, as can be observed in the SPSS table entitled 'Tests of Within-Subjects Contrasts', there was both a statistically significant linear, $F(1, 19) = 20.37$, $p < .001$, partial η^2 = .517, and quadratic, $F(1, 19) = 5.44$, $p = .031$, partial η^2 = .223, effect in the mean ranks.

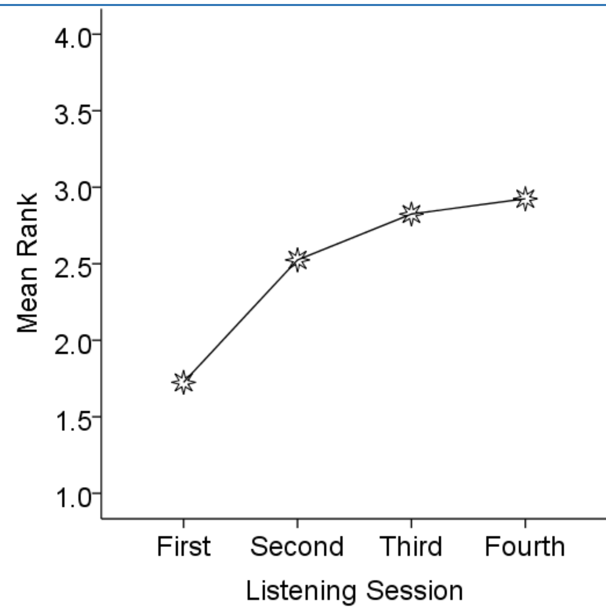
Tests of Within-Subjects Contrasts

Measure: MEASURE_1

Source	factor1	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Linear	15.210	1	15.210	20.366	.000	.517
	Quadratic	2.450	1	2.450	5.444	.031	.223
	Cubic	.090	1	.090	.379	.545	.020
Error(factor1)	Linear	14.190	19	.747			
	Quadratic	8.550	19	.450			
	Cubic	4.510	19	.237			

A visual appreciation of the linear and quadratic effect can be appreciated by viewing the plot of the mean ranks depicted in the Figure C16.3. If you are really keen, you could add error bars to the mean ranks with the same procedure I described in chapter 8.

Figure C16.3. Plot of the Mean Ranks



Trend Analysis on Ranks

A much more powerful approach is to conduct a linear trend analysis. A linear trend analysis would test the hypothesis that there is a linear increasing (or decreasing) trend in the mean ranks across the levels of the independent variable. Unfortunately, it is not possible to test such a hypothesis within the context of Friedman's ANOVA. However, if the data were ranked manually by the user, and then a regular one-way within-subjects ANOVA were performed on the ranks, a trend analysis could be performed. I performed such an analysis in SPSS.

As can be seen in the SPSS table entitled 'Tests of Within-Subjects Effects', the null hypothesis of equal mean ranks was rejected, $F(5, 350) = 9.76$, $p < .001$, partial $\eta^2 = .122$ (**Watch Video 16.23: Trend Analysis on Ranks**). Thus, the ANOVA on ranks yielded the same estimate of partial η^2 as that obtained from the Friedman's ANOVA/Kendall's W (i.e., .122):

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Tests of Within-Subjects Effects

Measure: MEASURE_1

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
offence	Sphericity Assumed	151.268	5	30.254	9.757	.000	.122
	Greenhouse-Geisser	151.268	4.405	34.341	9.757	.000	.122
	Huynh-Feldt	151.268	4.737	31.931	9.757	.000	.122
	Lower-bound	151.268	1.000	151.268	9.757	.003	.122
Error(offence)	Sphericity Assumed	1085.232	350	3.101			
	Greenhouse-Geisser	1085.232	308.342	3.520			
	Huynh-Feldt	1085.232	331.614	3.273			
	Lower-bound	1085.232	70.000	15.503			

Additionally, as can be seen in the table entitled 'Tests of Within-Subjects Contrasts', the test of a linear trend was statistically significant, $F(1, 70) = 37.57$, $p < .001$, partial $\eta^2 = .349$. The quadratic effect was not quite statistically significant, however, $F(1, 70) = 3.22$, $p = .077$, partial $\eta^2 = .044$. Had the quadratic effect been statistically significant, it would be necessary to plot the means (i.e., mean ranks) to appreciate the nature of the effect.

Tests of Within-Subjects Contrasts

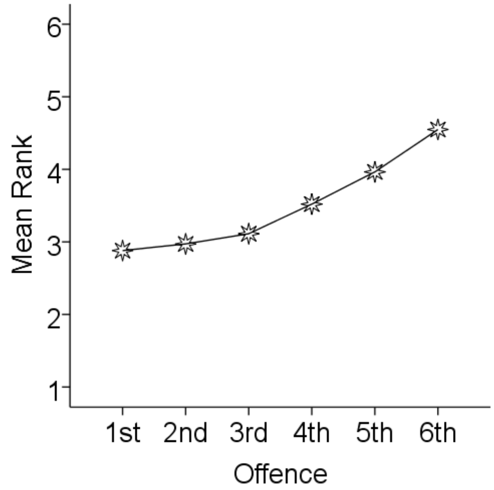
Measure: MEASURE_1

Source	offence	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
offence	Linear	139.615	1	139.615	37.568	.000	.349
	Quadratic	11.422	1	11.422	3.224	.077	.044
	Cubic	.023	1	.023	.009	.925	.000
	Order 4	.032	1	.032	.010	.920	.000
	Order 5	.175	1	.175	.068	.794	.001
Error(offence)	Linear	260.142	70	3.716			
	Quadratic	247.971	70	3.542			
	Cubic	176.494	70	2.521			
	Order 4	221.325	70	3.162			
	Order 5	179.301	70	2.561			

As can be seen Figure C16.4, there was principally a linear increase in the mean ranks across offences; however, there was also the suggestion of a bend in the trend such that the change from the third to the fourth offence was steeper than the changes between the earlier offences (hence the significant quadratic effect). Again, the quadratic effect was not statistically significant, however ($p = .077$).

Gignac, G. E. (2023). *How2statsbook* (Online Edition 2). Perth, Australia: Author.

Figure C16.4: Plot of Mean Ranks Across Offence



One-Way Within-Subjects ANOVA: Bootstrapping?

If you look at the plot of the means in Figure C16.4, you may appreciate that it was not it is not ideal, as the *Y*-axis represents mean ranks – values which are not naturally interpretable. This is an unfortunate fact, given that the dependent variable was measured with an easily interpreted scale: kilometers travelled. Consequently, my personal preference would be to conduct a regular one-way within subjects ANOVA on these data, so that I could interpret the results as relevant to kilometers, not ranks.

Ideally, a bootstrapped approach to conducting the one-way within-subjects ANOVA would be available in SPSS, because bootstrapping has been shown to accommodate data that are skewed up to|3.0|, even when the data are associated with moderate levels of violation of sphericity (Berkovits, Hancock, & Nevitt, 2000). Unfortunately, bootstrapping for a one-way within-subjects ANOVA is not available in SPSS. I suppose if I were very keen on sticking with means expressed in kilometers in this investigation, I would apply a regular one-way within-subjects ANOVA with the Huynh-Feldt adjustment, because Berkovits et al. (2000) found that it kept the error rate close to .05, when data were skewed up to |3.0| and the sample size was at least 60.

Practice Questions

"Quitting smoking is easy. I've done it thousands of times." Mark Twain

95% of attempts to quit smoking fail (Hughes, Keely, & Naud, 2004). Free et al. (2015) sought to improve the success rate, by sending text message support to smokers trying to quit. An example text message included: "Cravings last less than 5 minutes on average. To help distract yourself, try sipping a drink slowly until the craving is over". The percentage of participants who quit smoking was measured at three time periods: Day 1 (pre-intervention), 4 weeks, and 6 months. Test the null hypothesis that the percentage of non-smokers was equal across the three time periods. (.04%, 28.7%, 19.8%.)

Gignac, G. E. (2023). *How2statsbook* (Online Edition 2). Perth, Australia: Author.

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